THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS FULL SYLLABUS PAPER SET 03

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The distance of the point $(-7, 4)$ from the x-axis is: 1

(a) 11 units **(b)** 4 units **(c)** 7 units **(d)** −4 units

2. Which of the following points does not lie in Quadrant II? 1

(a) $(4, -4)$ **(b)** $(-7, 2)$ **(c)** $(-4, 4)$ **(d)** $(-1, 9)$

3. The point $(-6, -2)$ lies: 1

(a) Quadrant II **(b)** Quadrant IV **(c)** Quadrant I **(d)** Quadrant III

4. The degree of the polynomial -9 is: 1

(a) 1 **(b)** 0 **(c)** 3 **(d)** 2

5. The lines $y = 3x + 2$ and $y = 3x - 5$ are: 1

(a) perpendicular **(b)** parallel

(c) coincident **(d)** intersecting at the origin

6. The number $\sqrt{12}$ is: 1

(a) rational **(b)** irrational

(c) neither rational nor irrational **(d)** not a real number

7. The set of all rational numbers is denoted by: 1

(a) $\mathbb{Q}$ **(b)** $\mathbb{N}$ **(c)** $\mathbb{Z}$ **(d)** $\mathbb{R}$

8. The value of $(-20) \div 4$ is: 1

(a) -5 **(b)** -3 **(c)** 5 **(d)** 4

9. $(x + y + z)^2$ contains how many terms when fully expanded (no like terms)? 1

(a) 3 **(b)** 6 **(c)** 9 **(d)** 4

10. Expand $(5x - 1)^2$: 1

(a) $25x^2 - 5x + 1$ **(b)** $25x^2 - 10x + 1$ **(c)** $25x^2 + 10x + 1$ **(d)** $25x^2 - 1$

11. In a circle of radius 10 cm, a chord is 6 cm from the centre. The length of the chord is: 1

(a) 8 cm **(b)** 18 cm **(c)** 14 cm **(d)** 16 cm

12. A chord of length 48 cm is at a distance of 7 cm from the centre of a circle. The radius of the circle is: 1

(a) 25 cm **(b)** 31 cm **(c)** 26 cm **(d)** 48 cm

13. An arc subtends an angle of 70° at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 145° **(b)** 140° **(c)** 70° **(d)** 35°

14. Taking $\pi = \frac{22}{7}$, the area of a circle of radius 7 cm is: 1

(a) 154 cm^2 **(b)** 44 cm^2 **(c)** 308 cm^2 **(d)** 77 cm^2

15. The length of an arc subtending 90° at the centre of a circle of radius 21 cm ($\pi = \frac{22}{7}$) is: 1

(a) 66 cm **(b)** 16.5 cm **(c)** 132 cm **(d)** 33 cm

16. The area of a parallelogram with base 15 cm and height 8 cm is: 1

(a) 120 cm^2 **(b)** 46 cm^2 **(c)** 135 cm^2 **(d)** 60 cm^2

17. A fair die is rolled once. The probability of getting an odd prime number is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{4}$

18. A bag contains 10 red, 6 blue and 4 green balls. One ball is drawn at random. The probability that it is blue is: 1

(a) $\frac{2}{5}$ **(b)** $\frac{3}{10}$ **(c)** $\frac{3}{11}$ **(d)** $\frac{7}{10}$

19. The Virahānka–Fibonacci sequence was first written down in connection with: 1

(a) calendar design **(b)** the area of circles

(c) planetary motion **(d)** Prakrit metre and poetry

20. The sum of the first 50 natural numbers is: 1

(a) 1274 **(b)** 2550 **(c)** 1275 **(d)** 1276

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. The midpoint of AB is $M(2, -1)$ and $A = (-4, 3)$. Find B. 2

22. Show that $(7, 15)$ lies on the line $y = 2x + 1$ but $(3, 8)$ does not. 2

23. Show that $\frac{4}{6}$ and $\frac{10}{15}$ are equal rational numbers. 2

24. Factorise $x^2 - 7x + 12$ by splitting the middle term. 2

25. In a circle of radius 5 cm, one chord is 3 cm from the centre and another is 4 cm from the centre. Which is longer? Find both lengths. 2

26. Use Heron's formula to find the area of a triangle of sides 3 cm, 4 cm and 5 cm and verify with $\frac{1}{2}bh$. 2

27. A coin is tossed 200 times and shows heads 112 times. Find the relative frequency of heads and compare it with $\frac{1}{2}$. 2

28. Show that $2, 6, 18, 54, \ldots$ is a GP and write its nth term. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Find a point on the y-axis equidistant from $A(6, 5)$ and $B(-4, 3)$. 3

30. Sarita has a 500-page book and reads 20 pages a day. Write the pages left after d days and find the days needed to finish. 3

31. A tailor has 15 m of silk and a kurta needs 2 m. How many kurtas can be made and how much silk is left? Express the number of kurtas as a rational number. 3

32. A square has side $(2x + 3)$ units. Find its area and perimeter. 3

33. The minute hand of a clock is 14 cm long. Find the distance its tip travels in 30 minutes and the area swept in that time ($\pi = \frac{22}{7}$). 3

34. A die is rolled once. Find the probability that the number is both even and prime, and the probability that it is neither. 3

35. A chord of a circle of radius r is at distance d from the centre. Derive the chord length. Hence find it for $r = 10$ and $d = 6$. 3

OR

Show that a cyclic parallelogram is a rectangle. 3

36. Find the first five terms of $u_1 = 1,\ u_n = 2u_{n-1} + 3$ and decide whether 133 is a term. 3

OR

A ball dropped from 24 ft rebounds to $\frac{3}{4}$ of its previous height each time. Find the heights after the first three bounces and state the type of sequence. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** Brahmagupta's rules: fortune + fortune = fortune; debt + debt = debt; debt × fortune = debt; debt × debt = fortune. Using these, 5

(i) compute $(-3) \times 4$ and $(-3) \times (-4)$;

(ii) a spice trader borrows ₹850, earns ₹1,200 and loses ₹450 — find his final standing;

(iii) explain $(-3) \times (-4) = 12$ in terms of debt.

OR

(i) Show that between any two rational numbers there is another rational number. 5

(ii) Find the rational number midway between $-\frac{1}{4}$ and $\frac{5}{12}$ and place it on the number line (describe).

38. **Case study:** A ground is a circle of radius 13 m. Two parallel boundary ropes (chords) are 10 m and 24 m long. 5

(i) Find the distance of each from the centre.

(ii) How far apart are they if on opposite sides of the centre?

(iii) How far if on the same side?

(iv) Which rope is longer and which is nearer to the centre?

OR

Prove that the angle subtended by an arc at a point on the circle is half the angle at the centre in the case where the line DC produced meets the circle at a point E outside the arc. 5

39. Baudhāyana's problem: 5

(i) Explain what it means to ‘square a rectangle’.

(ii) Find the side of the square equal in area to a $25\ \text{cm} \times 4\ \text{cm}$ rectangle.

(iii) Find its perimeter and compare it with the rectangle's perimeter.

OR

Case study: A cyclic plot has sides 25 m, 39 m, 52 m and 60 m. 5

(i) Find its semi-perimeter.

(ii) Find its area by Brahmagupta's formula.

(iii) Show that the plot's diagonal of length 65 m is a diameter by finding the area as two right triangles (25-60-65 and 39-52-65).

(iv) Find the cost of levelling at ₹20 per m^2.

40. **Case study:** Meera saves ₹100 in the first month and ₹50 more each month than in the previous month. 5

(i) Write her savings for the first three months.

(ii) Find the nth term.

(iii) Find her saving in the 12th month and her total in a year.

(iv) In which month does she save ₹1,000?

OR

Case study: A colony of 500 bacteria doubles every hour. 5

(i) Write the numbers at the end of hours 1 to 4.

(ii) Write the nth term.

(iii) How many bacteria after 6 hours?

(iv) After how many hours does the number first exceed 10,000?

END OF QUESTION PAPER **Total: 80 Marks**

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